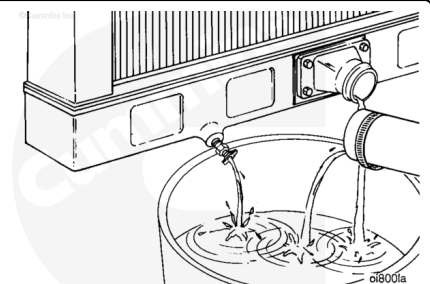


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

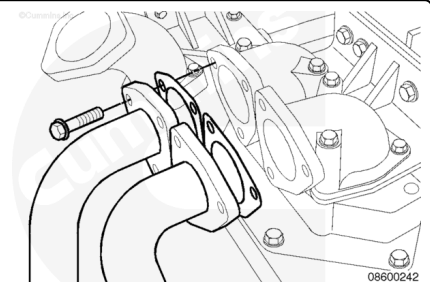
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)

Remove

It is a good practice to mark the location of all lines during removal.

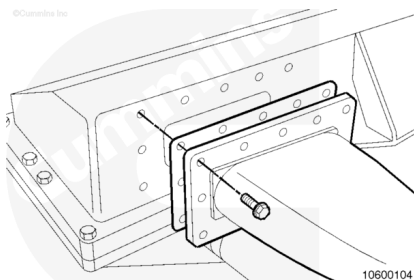
Disconnect the coolant lines from the thermostat housing.

Position coolant lines aside.

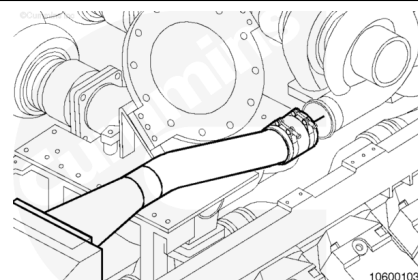


Remove the 12 capscrews from each of the four intercooler intake crossovers and the two intercoolers.

Remove and discard the gasket.



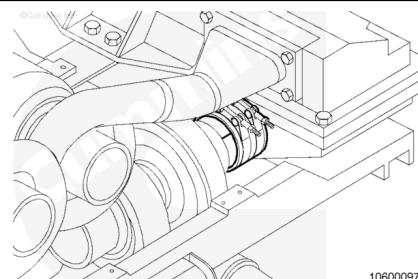
Loosen the hose clamps on each end of the four air connection hoses for the four intercooler intake crossovers and the four low-pressure turbochargers.



Note : The left rear high-pressure turbocharger connection is shown. The other high-pressure turbocharger connections are similar.

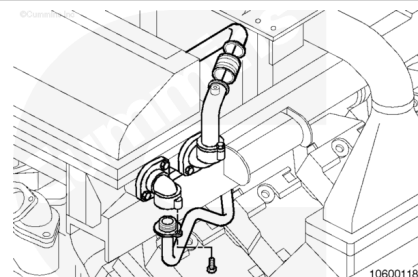
Loosen the two intake hose clamps on the four high-pressure turbochargers and the two intercoolers.

Disconnect the hose clamp at the intercooler.

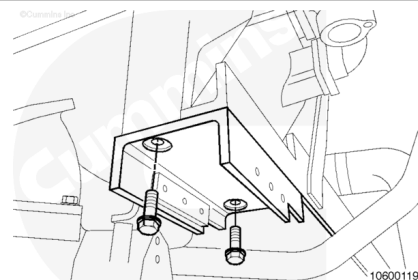


Note : The front intercooler water supply and return connections are shown. The rear intercooler connections are similar.

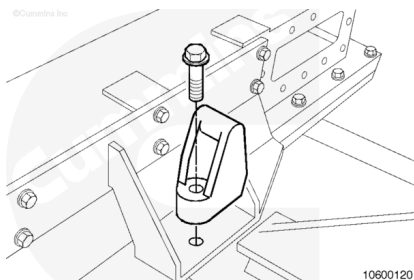
Disconnect the water supply and return piping from the intercoolers.



Remove the capscrews from each of the four intercooler supports.



Remove the lifting eye from the bracket between the intercooler and the two-stage frame.



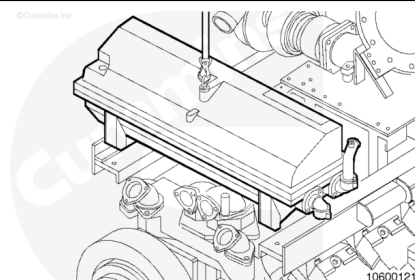
10600120

⚠ WARNING ⚠

The QSV81G and QSV91G engines can weigh as much as 11,000 kg [24,200 lbs]. Use a properly rated hoist and engine lifting fixture, Part Number 3163264, to lift the engine. Rigging and lifting must be done by trained, experienced personnel.

Install a M12 x 1.75 threaded eye bolt in the top of the intercoolers.

Remove the intercoolers.

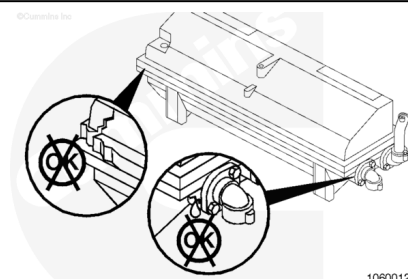


10600121

Inspect for Reuse

Note : After an instance of turbocharger malfunction, inspect thoroughly for debris throughout the system.

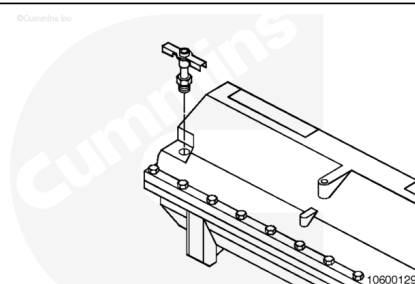
Inspect the intercooler assembly for cracks and indications of coolant leakage.



10600128

Disassemble

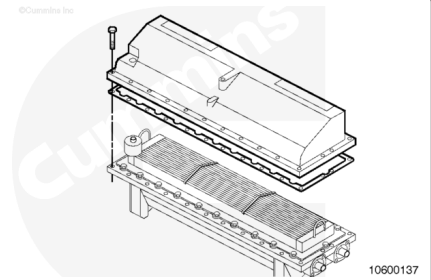
Remove the petcock vent from the intercooler.



10600129

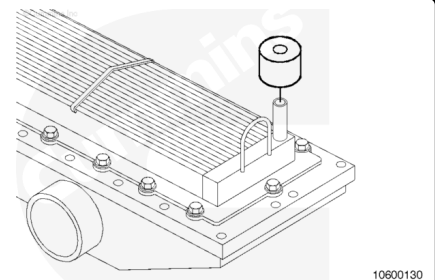
⚠ WARNING ⚠

The QSV81G and QSV91G engines can weigh as much as 11,000 kg [24,200 lbs]. Use a properly rated hoist and engine lifting fixture, Part Number 3163264, to lift the engine. Rigging and lifting must be done by trained, experienced personnel.

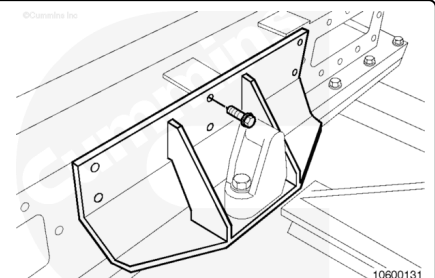


Remove the 26 capscrews, intercooler cover, and intercooler gasket from the intercooler housing.

Remove the air grommet from the air vent tube.

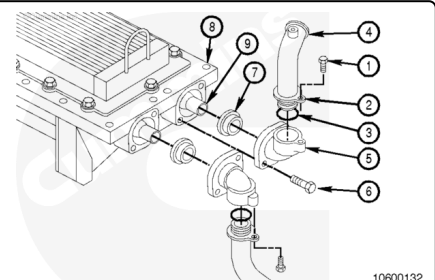


Remove the intercooler mounting bracket.

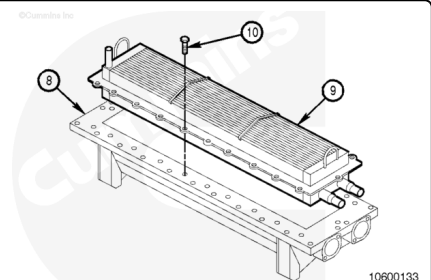


Remove capscrew (1), flange (2), pipe (4) and seal (3) from each of the intercooler fittings (5).

Remove the capscrews (6), intercooler fittings (5), and seals (7) from the intercooler housing (8) and intercooler element (9).



⚠ WARNING ⚠

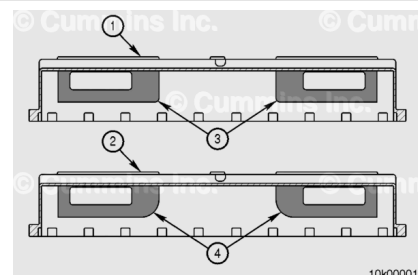


The QSV81G and QSV91G engines can weigh as much as 11,000 kg [24,200 lbs]. Use a properly rated hoist and engine lifting fixture, Part Number 3163264, to lift the engine. Rigging and lifting must be done by trained, experienced personnel.

Remove the 22 capscrews (10) and the intercooler element (9) from the intercooler housing.

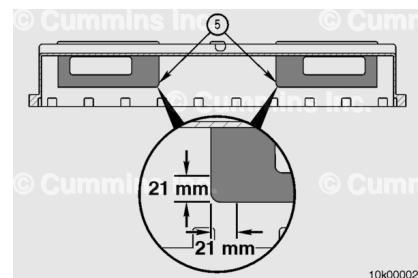
Machine

A new intercooler element has been released, which if, fitted with older style intercooler covers, causes an interference condition. The older (1) and new (2) style covers can be easily identified by the inner bossess on the crossover flange as shown. The older covers have a square corner (3) compared to the more rounded corner of the new covers (4)



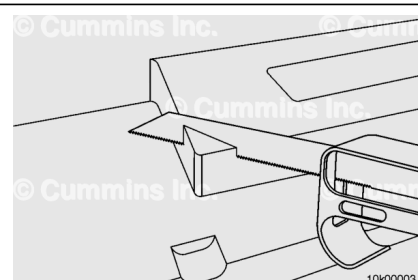
⚠ WARNING ⚠

The QSV81G and QSV91G engines can weigh as much as 11,000 kg [24,200 lbs]. Use a properly rated hoist and engine lifting fixture, Part Number 3163264, to lift the engine. Rigging and lifting must be done by trained, experienced personnel.

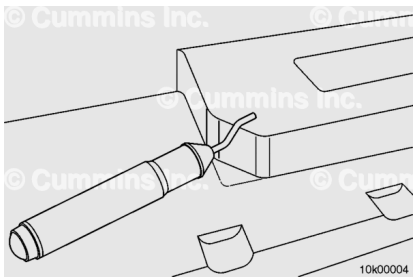


In order to make the older style intercooler cover fit with the new element some material will need to be removed from the cover. First mark out the area to be removed. From the corner of the boss marked (5), measure 21mm [0.83 in] along both the x and y axis. Join these points together. Repeat for the adjacent boss as shown in the illustration.

Use an appropriate cutting tool to carefully remove the material that has been marked. Cut along the horizontal and verical planes to remove the material. Make sure that the cut does **not** go beyond the face at the bottom of the boss.



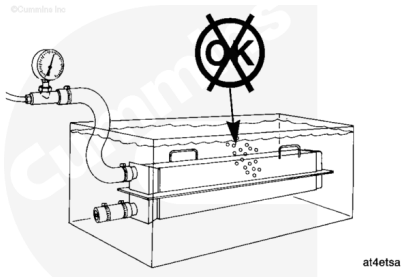
Use an appropriate deburring tool to remove any sharp edges from the machined area.



Pressure Test

⚠ WARNING ⚠

The QSV81G and QSV91G engines can weigh as much as 11,000 kg [24,200 lbs]. Use a properly rated hoist and engine lifting fixture, Part Number 3163264, to lift the engine. Rigging and lifting must be done by trained, experienced personnel.



Heating the tank water to 50°C [122°F] improves the test conditions.

Plug one intercooler water tube with a hose that has a pipe plug secured into one end.

Attach an air line and a regulator to the remaining water tube.

Apply air pressure and submerge the intercooler element into a tank of water.

Measurements

	kpa	psi
Air Pressure	415	60

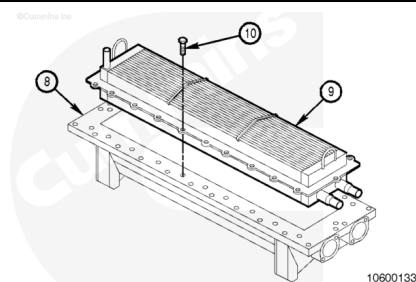
Inspect the intercooler element for leaks.

Do **not** attempt to repair a leaking intercooler element. If an intercooler element has a leak, it **must** be replaced.

Assemble

⚠ WARNING ⚠

The QSV81G and QSV91G engines can weigh as much as 11,000 kg [24,200 lbs]. Use a properly rated hoist and engine lifting fixture, Part Number 3163264, to lift the engine. Rigging and lifting must be done by trained, experienced personnel.

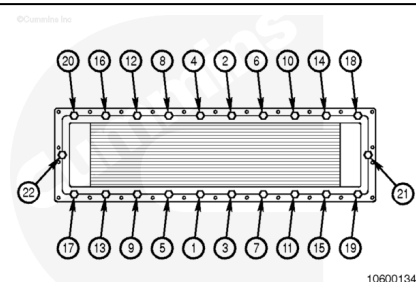


The intercooler element **must** be positioned on the intercooler housing alignment dowels during installation.

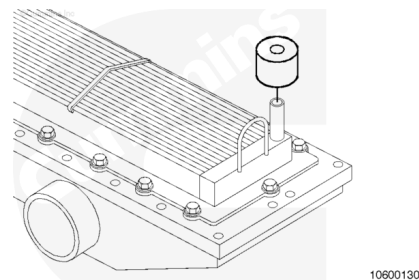
Position the intercooler element (9) in the intercooler housing (8) and install the 22 capscrews (10).

Tighten the capscrews in the sequence shown in the illustration.

Torque Value: 45 n•m [33 ft-lb]



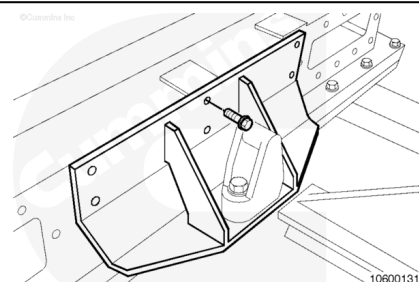
Install the air grommet on the air vent tube.



Install the intercooler bracket and six capscrews.

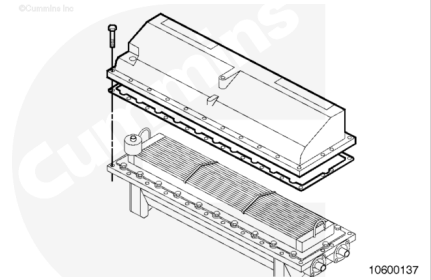
Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



⚠ WARNING ⚠

The QSV81G and QSV91G engines can weigh as much as 11,000 kg [24,200 lbs]. Use a properly rated hoist and engine lifting fixture, Part Number 3163264, to lift the engine. Rigging and lifting must be done by trained, experienced personnel.



Install a 12 mm [0.47 in] lifting device in the top of the intercooler cover.

Position the intercooler gasket and cover on the intercooler.

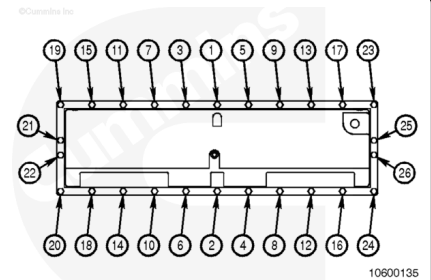
Install the 26 intercooler cover capscrews.

Slightly tighten the four capscrews near the air grommet to hold the air grommet in position.

Note : If installing a new style intercooler cover and there is interference condition, reference the Machine Section in this procedure. Check for characteristics and detail.

Tighten the capscrews in the sequence shown in the illustration.

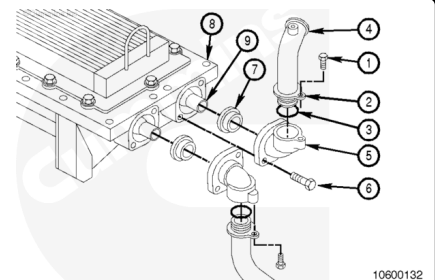
Torque Value: 45 n·m [33 ft-lb]



Install the flange (2), seal (3), and pipe (4) in each of the two intercooler fittings (5) with the capscrew (1).

Install the two intercooler fittings (5) and seals (7) in the intercooler (8) and element (9) with the capscrews (6).

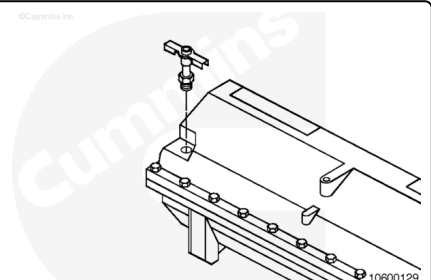
Torque Value: 45 n·m [33 ft-lb]



Apply pipe sealant to the threads of the petcock.

Install the petcock vent on the intercooler.

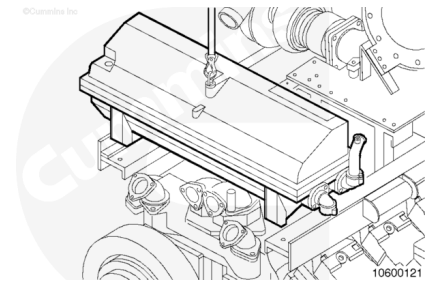
Torque Value: 8 n·m [71 in-lb]



Install

⚠ WARNING ⚠

The QSV81G and QSV91G engines can weigh as much as 11,000 kg [24,200 lbs]. Use a properly rated hoist and engine lifting fixture, Part Number 3163264, to lift the engine. Rigging and lifting must be done by trained, experienced personnel.



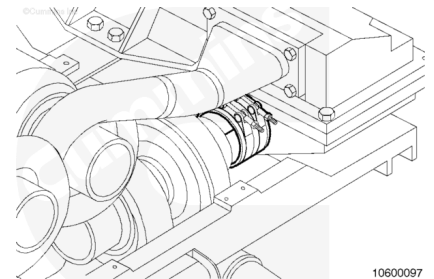
Install the lifting eye to the top of the intercoolers.

Make sure the hoses from the intercooler to the high-pressure turbochargers are aligned during installation.

Position the intercooler on the frame.

⚠ CAUTION ⚠

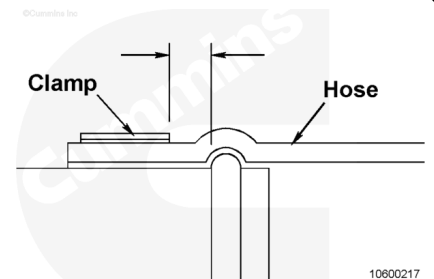
The intercooler air hoses contain glass fibers. Installing a hose without first moving the intercooler, and/or forcing the hose into position will damage the glass fibers and cause premature failure.



Install the intake hoses and two clamps each on the four high-pressure turbochargers.

⚠ CAUTION ⚠

Do not overtighten the clamps. Overtightening the clamps will damage the hose and cause premature failure.



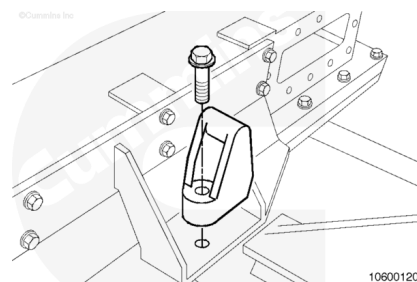
The clamps **must** be positioned so that the edge of the clamp is between 3 mm [0.125 in] and 6 mm [0.250 in] from the inner edge of the bead. Clamps **must not** overhang the edge of the hose.

Torque Value: 9 n•m [80 in-lb]

Install the intercooler lifting eye from the intercooler housing.

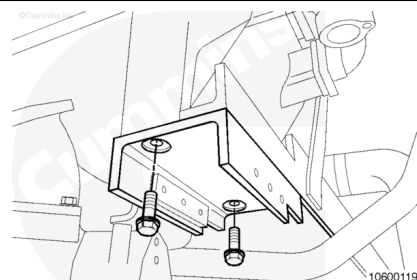
Install the lifting eye mounting capscrews onto the two-stage frame.

Torque Value: 210 n•m [155 ft-lb]



Install the intercooler mounting capscrews at each of the four supports.

Torque Value: 210 n•m [155 ft-lb]

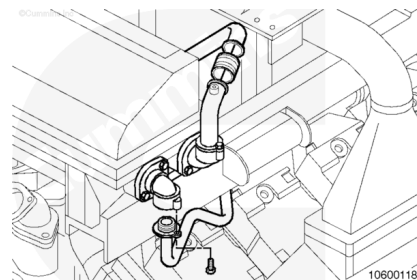


Position new o-rings on the coolant supply and return piping.

Install the coolant supply and return piping into the intercoolers.

Install and tighten the flange capscrews if removed.

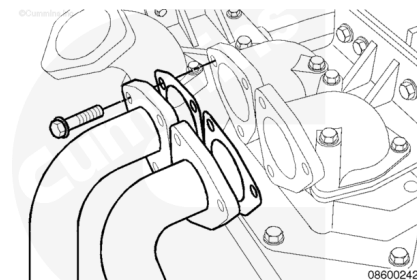
Torque Value: 45 n•m [33 ft-lb]



Install the LTA thermostat housing coolant supply lines.

Tighten the capscrews.

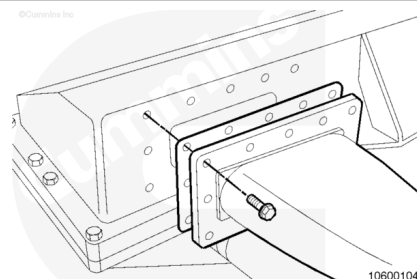
Torque Value: 54 n•m [40 ft-lb]



Install the four intercooler intake crossovers with 12 capscrews at each crossover.

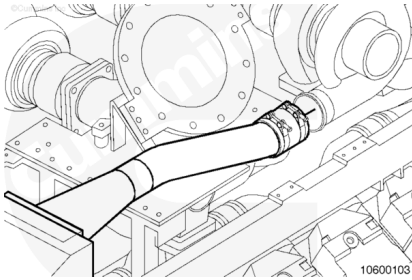
Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



Install the hoses and clamps on the four intercooler intake crossovers and the low-pressure turbochargers.

Torque Value: 9 n•m [80 in-lb]



Finishing Steps

- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>)
- Operate the engine and check for leaks.

